Northfield Institute of Technology

Course Plan: Database Management Systems (21CS53), V Semester

Faculty: Prof. S. Rao Credits: 4 Academic year 2026-27

Units

Unit	Topics	Hours
1	Introduction, data models, DBMS architecture	8
2	Relational model, relational algebra	8
3	SQL: queries, joins, views, triggers	8
4	ER modelling, ER-to-relational mapping, normalisation (1NF to BCNF)	8
5	Transactions, concurrency control, recovery	8

Assessment

• Assignment 1 covers Units 1-2. **Assignment 2 covers Unit 4 (ER modelling and normalisation)** and must include at least two complete ER diagrams converted to relational schemas.

• Assignments are submitted on the college portal only. Late submissions are not accepted.

• Internal Assessment tests: IA-1 (Units 1-2), IA-2 (Units 3-4).

Laboratory

• The lab component has 8 experiments. **All 8 experiments must be signed by the faculty in the lab record before the lab viva.** Students without a fully signed record will not be allowed to take the viva.

• Record signing: Prof. Rao signs records in the department office on Tuesdays and Wednesdays, 3:00-5:00 PM.

• Lab viva carries 10 marks of the 25 internal lab marks.